Conceptual Notes

The Geometry of Minkowski Spacetime

G. L. Naber (2nd ed., Springer 2011)

Introduction & Sections 1.1–1.2

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Preface Remarks (Summary)

The **Introduction** motivates the mathematical model of "the event world" via Zeeman's 1964 theorem. Chapter 1 (first five sections) develops the basic geometry of this model, including indefinite inner products, orthonormal bases, and the Lorentz group. Section 1.6 of Chapter 1 gives a complete proof of Zeeman's theorem.

1 Introduction

1.1 The Goal: Modelling the Event World

The **fundamental goal** of the book is to understand the mathematical structure that physics imposes on the set of all events.

Definition 1.1.1: Event (Point-Event)

An **event** is an idealized physical occurrence that has *no spatial extension* and *no duration in time* — a "point-event." Examples: an instantaneous collision, an instant in the history of a material particle or photon.
The **worldline** of a particle or photon is the continuous sequence of events that constitutes its history.
We begin with an abstract set $\mathcal{M}$ whose elements are called events.

1.2 Admissible Observers and Frames of Reference

Definition 1.2.1: Admissible Observer

An **admissible observer** presides over a *3-dimensional, right-handed, Cartesian spatial coordinate system* based on an agreed unit of length, relative to which photons propagate rectilinearly in any direction.
Notation: observers $O, \hat{O}, \ldots$ preside over spatial coordinate systems $\Sigma(x^1, x^2, x^3)$, $\hat{\Sigma}(\hat{x}^1, \hat{x}^2, \hat{x}^3), \ldots$

Remarks on admissibility.

(i) "Presides over" does not mean the observer is ubiquitous; the observer sits at the origin and receives communicated information about distant events.

(ii) The rectilinear propagation condition is a genuine restriction: it excludes rotating coordinate systems (where light would *not* travel in straight lines).

(iii) This does *not* preclude two admissible observers being in relative motion.

Definition 1.2.2: Ideal Standard Clock

Each admissible observer is equipped with an **ideal standard clock** (based on an agreed unit of time) to provide a quantitative temporal order to events on their worldline.

1.2.1 The Problem of Moving Clocks

To assign times to *all* events (not just those on the observer's worldline), one must synchronize clocks throughout the coordinate system. A natural attempt—moving clocks from the

origin—fails:

Empirical fact (moving clocks). Two identical atomic clocks manufactured and synchronized in New York: one stays put, one is flown around the world. Upon return, the travelling clock *lags* behind the stationary one. Moving clocks do not remain synchronized.

1.2.2 Clock Synchronization via Light Signals

The resolution: build clocks *in place*, then synchronize using light signals.

Definition 1.2.3: Fizeau Procedure

An observer at origin O emits a light signal at clock-time t_0. The signal reflects from a point P and returns at time t_1. The speed of the signal is computed as:

$$c = \frac{\text{distance}(O, P)}{\frac{1}{2}(t_1 - t_0)}.$$

(The factor $\frac{1}{2}$ accounts for the round trip; no synchronized clock at P is needed.)

Key Principle / Postulate 1.2.1: Constancy and Universality of the Speed of Light

For each admissible observer, the speed of light in vacuo as measured by the Fizeau procedure is:

(i) independent of *when* the experiment is performed;
(ii) independent of the *arrangement of apparatus* (choice of P);
(iii) independent of the *frequency (energy)* of the signal;
(iv) **the same numerical value $c \approx 3.0 \times 10^8$ m/s for all admissible observers.**

This is the consolidated conclusion of Michelson-Morley, Kennedy-Thorndike, and many other experiments.

Definition 1.2.4: Units: Time Measured in Distance (Light Travel Time)

Each observer multiplies time readings by c, so that time is measured in units of length (e.g., "one meter of time" = time for light to travel one meter). Under these units all speeds are dimensionless and $c = 1$.
Time coordinates for observers $O, \hat{O}, \ldots$ are denoted $x^4 = ct,\ \hat{x}^4 = c\hat{t},\ \ldots$

Definition 1.2.5: Synchronization Procedure

At each point P of the spatial coordinate system, place an identical clock. At time x^4 (at origin O), emit a spherical electromagnetic wave. When the wavefront reaches P,

set the clock there to

$$x^4 + \text{distance}(O, P)$$

and start it ticking. Result: all clocks synchronized with the origin clock.

Definition 1.2.6: Frame of Reference

Each admissible observer O has thereby established a **frame of reference** $S(x^1, x^2, x^3, x^4)$. Intuitive picture: a *lattice of spatial coordinate lines* with at each lattice point a synchronized clock and an assistant who records locations and times of nearby events.

1.3 The Coordinate Transformation Problem

Question: How are the $\hat{S}$-coordinates $(\hat{x}^1, \hat{x}^2, \hat{x}^3, \hat{x}^4)$ of an event related to its S-coordinates (x^1, x^2, x^3, x^4)?

Define the **coordinate transformation map**

$$F : \mathbb{R}^4 \to \mathbb{R}^4, \qquad F(x^1, x^2, x^3, x^4) = (\hat{x}^1, \hat{x}^2, \hat{x}^3, \hat{x}^4).$$

F must be bijective (its inverse F^{-1} transforms from hatted to unhatted coordinates).

To determine more structure we need an additional postulate.

Key Principle / Postulate 1.3.1: Causality Assumption (Temporal Order of Photon Events)

Any two admissible observers **agree on the temporal order** of any two events on the worldline of a photon.
That is, if two photon-worldline events have S-coordinates (x^1, x^2, x^3, x^4) and $(x_0^1, x_0^2, x_0^3, x_0^4)$, and $\hat{S}$-coordinates $(\hat{x}^1, \hat{x}^2, \hat{x}^3, \hat{x}^4)$ and $(\hat{x}_0^1, \hat{x}_0^2, \hat{x}_0^3, \hat{x}_0^4)$, then $x^4 - x_0^4$ and $\hat{x}^4 - \hat{x}_0^4$ have the *same sign*.
Note: We only assert same sign (same temporal order), *not* equal magnitudes.

1.3.1 The Light Cone Condition

Since photons travel rectilinearly with speed $c = 1$, two events on a photon worldline satisfy in S:

$$x^i - x_0^i = v^i\,(x^4 - x_0^4), \quad i = 1, 2, 3, \qquad (v^1)^2 + (v^2)^2 + (v^3)^2 = 1,$$

which gives the **null cone condition**:

$$(x^1 - x_0^1)^2 + (x^2 - x_0^2)^2 + (x^3 - x_0^3)^2 - (x^4 - x_0^4)^2 = 0. \tag{0.1}$$

Geometrically, (0.1) is a *cone in* $\mathbb{R}^4$ with vertex at $(x_0^1, x_0^2, x_0^3, x_0^4)$ (analogous to the right-circular cone $(z - z_0)^2 = (x - x_0)^2 + (y - y_0)^2$ in $\mathbb{R}^3$).

The same condition holds in $\hat{S}$:

$$(\hat{x}^1 - \hat{x}^1_0)^2 + (\hat{x}^2 - \hat{x}^2_0)^2 + (\hat{x}^3 - \hat{x}^3_0)^2 - (\hat{x}^4 - \hat{x}^4_0)^2 = 0. \tag{0.2}$$

Consequence. The map F must carry cone (0.1) to cone (0.2), and must satisfy $\hat{x}^4 > \hat{x}^4_0$ whenever $x^4 > x^4_0$ and (0.1) holds.

1.4 Zeeman's Theorem and the Three Types of Causal Automorphisms

Definition 1.4.1: Causal Automorphism

A bijection $F : \mathbb{R}^4 \to \mathbb{R}^4$ is a **causal automorphism** if:
(i) F carries cone (0.1) onto cone (0.2), and
(ii) $\hat{x}^4 > \hat{x}^4_0$ whenever $x^4 > x^4_0$ and (0.1) holds.

Theorem 1.4.1: Zeeman's Theorem (1964)

Every causal automorphism $F : \mathbb{R}^4 \to \mathbb{R}^4$ is a composition of maps of the following three basic types:

1. **Translations:** $\hat{x}^a = x^a + \Lambda^a$, $a = 1, 2, 3, 4$, for constants Λ^a.
2. **Positive scalar multiples:** $\hat{x}^a = kx^a$, $a = 1, 2, 3, 4$, for some positive constant k.
3. **Orthochronous Lorentz transformations (linear):**

$$\hat{x}^a = \Lambda^a{}_b\, x^b, \qquad a = 1, 2, 3, 4, \tag{0.3}$$

where $\Lambda = [\Lambda^a{}_b]_{a,b=1,2,3,4}$ satisfies

$$\Lambda^T \eta \Lambda = \eta, \tag{0.4}$$

with

$$\eta = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix},$$

and

$$\Lambda^4{}_4 \geq 1. \tag{0.5}$$

Remarkable feature: F is not assumed to be continuous, much less linear — yet the conclusion is this strong. (Proof given in Section 1.6.)

1.5 Reduction to Orthochronous Lorentz Transformations

Eliminating Types 1 and 2.

- Type 2 (scaling): Two frames related by a positive scalar multiple differ only by a trivial change of scale. We exclude these.
- Type 1 (translations): The constants Λ^a are merely the $\hat{S}$-coordinates of S's spacetime origin. We require all observers to agree on a common origin event, so $\Lambda^a = 0$ for $a = 1, 2, 3, 4$.

What remains are exactly the transformations of the form (0.3) satisfying (0.4) and (0.5): the **orthochronous Lorentz transformations.**

Structural analogy. Orthochronous Lorentz transformations play the same role for the quadratic form $(x^1)^2 + (x^2)^2 + (x^3)^2 - (x^4)^2$ as orthogonal transformations play for the Euclidean squared length $x^2+y^2+z^2$ in $\mathbb{R}^3$. They are precisely the linear transformations that leave this quadratic form invariant and preserve time orientation ($\Lambda^4{}_4 \geq 1$).

1.6 The Relativity Principle

Key Principle / Postulate 1.6.1: The Relativity Principle

All admissible frames of reference are completely equivalent for the formulation of the laws of physics.

Conceptual content: No admissible observer has a privileged view. None can legitimately claim to be "at rest" while the others "move." All are simply in relative motion. In particular, admissible observers can disagree about the temporal order of spatially separated events ("simultaneity"), and the Relativity Principle forbids preferring any one observer's judgment.

Caveat: Idealization. No exact admissible frame of reference exists physically. The notion is an idealization. Gravitational fields impose severe restrictions on the spatial and temporal extent of any admissible frame. We restrict to regimes where gravitational effects are negligible (until Chapter 4).

2 Chapter 1: Geometrical Structure of $\mathcal{M}$

2.1 Section 1.1: Preliminaries

2.1.1 Bilinear Forms and Inner Products

Let V denote an arbitrary real vector space of dimension $n \geq 1$.

Definition 2.1.1: Bilinear Form

A **bilinear form** on V is a map $g : V \times V \to \mathbb{R}$ that is linear in each variable:

$$g(a_1 v_1 + a_2 v_2,\, w) = a_1\, g(v_1, w) + a_2\, g(v_2, w),$$

$$g(v,\, a_1 w_1 + a_2 w_2) = a_1\, g(v, w_1) + a_2\, g(v, w_2).$$

Definition 2.1.2: Symmetric, Nondegenerate, Inner Product

A bilinear form g is:

- **symmetric** if $g(w, v) = g(v, w)$ for all $v, w \in V$.
- **nondegenerate** if $g(v, w) = 0$ for all $w \in V$ implies $v = 0$.
- an **inner product** if it is both nondegenerate and symmetric.

Notation: $g(v, w)$ is often written $v \cdot w$.
Standard example: On $\mathbb{R}^n$, the usual inner product is $g(v, w) = v^1 w^1 + \cdots + v^n w^n$.
This is **positive definite**: $v \neq 0 \Rightarrow g(v, v) > 0$. Not all inner products share this.

Exercise 2.1.1: Exercise 1.1.1 — A Non-Positive-Definite Inner Product

Define $g_1 : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ by

$$g_1(v, w) = v^1 w^1 + v^2 w^2 + \cdots + v^{n-1} w^{n-1} - v^n w^n.$$

(a) Show that g_1 is an inner product.
(b) Exhibit nonzero vectors v, w such that $g_1(v, v) = 0$ and $g_1(w, w) < 0$.

Solution sketch.

- Linearity in each variable is clear from the definition.
- **Symmetry:** $g_1(v, w) = g_1(w, v)$ by commutativity of multiplication.
- **Nondegeneracy:** If $g_1(v, w) = 0$ for all w, take $w = e_i$ (standard basis vectors) to get $v^i = 0$ for $i < n$ and $v^n = 0$, so $v = 0$.
- For (b): take $v = (1, 0, \ldots, 0, 1)$ so $g_1(v, v) = 1 - 1 = 0$; take $w = (0, \ldots, 0, 1)$ so $g_1(w, w) = -1 < 0$.

Definition 2.1.3: Definite and Indefinite Inner Products

An inner product g on V is:

- **positive definite** if $v \neq 0 \Rightarrow g(v, v) > 0$.

- **negative definite** if $v \neq 0 \Rightarrow g(v,v) < 0$.
- **indefinite** if it is neither positive definite nor negative definite.

Definition 2.1.4: Orthogonality and Orthogonal Complement

Given an inner product g on V:
- v and w are **g-orthogonal** (or simply orthogonal) if $g(v,w) = 0$.
- The **orthogonal complement** of a subspace $W \subseteq V$ is

$$W^\perp = \{v \in V : g(v,w) = 0 \text{ for all } w \in W\}.$$

Exercise 2.1.2: Exercise 1.1.2 — $W^\perp$ is a Subspace

Show that $W^\perp$ is a subspace of V.

Solution. We verify the three subspace conditions:
(i) $g(0,w) = 0$ for all w, so $0 \in W^\perp$.
(ii) If $v_1, v_2 \in W^\perp$, then $g(v_1+v_2, w) = g(v_1,w) + g(v_2,w) = 0 + 0 = 0$, so $v_1 + v_2 \in W^\perp$.
(iii) If $v \in W^\perp$ and $a \in \mathbb{R}$, then $g(av,w) = a\,g(v,w) = 0$, so $av \in W^\perp$.

Definition 2.1.5: Quadratic Form

The **quadratic form** associated with g is the map $Q : V \to \mathbb{R}$,

$$Q(v) = g(v,v) = v \cdot v \quad \text{(sometimes written } v^2\text{)}.$$

Key fact: Distinct inner products give rise to distinct quadratic forms (see Exercise 1.1.3 below).

Exercise 2.1.3: Exercise 1.1.3 — Quadratic Form Determines the Inner Product

If g_1 and g_2 are two inner products on V satisfying $g_1(v,v) = g_2(v,v)$ for all $v \in V$, then $g_1(v,w) = g_2(v,w)$ for all $v, w \in V$.

Solution. Let $h = g_1 - g_2$, which is bilinear and symmetric with $h(v,v) = 0$ for all v. Evaluate:

$$0 = h(v+w, v+w) = h(v,v) + 2\,h(v,w) + h(w,w) = 0 + 2\,h(v,w) + 0,$$

so $h(v,w) = 0$ for all v, w, i.e., $g_1 = g_2$. ■

Definition 2.1.6: Unit Vectors and Orthonormal Bases

- A vector v is a **unit vector** if $Q(v) = \pm 1$.
- A basis $\{e_1, \ldots, e_n\}$ for V is an **orthonormal basis** if the e_i are mutually orthogonal unit vectors: $g(e_i, e_j) = 0$ for $i \neq j$, and $Q(e_i) = \pm 1$.

2.1.2 Sylvester's Theorem: Existence and Uniqueness of the Index

Theorem 2.1.1: Theorem 1.1.1 (Existence and Invariance of Orthonormal Bases)

Let V be an n-dimensional real vector space with a nondegenerate, symmetric, bilinear form g. Then:

1. There exists an orthonormal basis $\{e_1, \ldots, e_n\}$ for V (i.e., $g(e_i, e_j) = 0$ for $i \neq j$ and $Q(e_i) = \pm 1$ for each i).
2. The number of basis vectors e_i with $Q(e_i) = -1$ is the **same** for any such orthonormal basis.

Proof. **Step 1: There exists $u \in V$ with $Q(u) \neq 0$.**

Since g is nondegenerate, there exist v, w with $g(v, w) \neq 0$. If $Q(v) \neq 0$ or $Q(w) \neq 0$, done. Otherwise $Q(v) = Q(w) = 0$, and then

$$Q(v + w) = Q(v) + 2g(v, w) + Q(w) = 2g(v, w) \neq 0,$$

so take $u = v + w$.

Step 2: Base case $n = 1$.

Pick any $u \in V$ with $Q(u) \neq 0$. Set $e_1 = |Q(u)|^{-1/2}\, u$. Then $Q(e_1) = \pm 1$ and $\{e_1\}$ is the required basis.

Step 3: Inductive step ($n > 1$).

Assume the result holds for all vector spaces of dimension $< n$. Pick $u \in V$ with $Q(u) \neq 0$ and set $e_n = |Q(u)|^{-1/2}\, u$, so $Q(e_n) = \pm 1$.

Let $W = (\mathrm{Span}\{e_n\})^{\perp}$. By Exercise 1.1.2, W is a subspace. Since $e_n \notin W$, we have $\dim W < n$. The restriction of g to $W \times W$ is an inner product on W (nondegeneracy of $g|_W$ follows since if $g(w, v) = 0$ for all $v \in W$ and also $g(w, e_n) = 0$, then $w = 0$). By the induction hypothesis, W has an orthonormal basis $\{e_1, \ldots, e_m\}$, $m = \dim W$.

Claim: $m = n - 1$ and $\{e_1, \ldots, e_m, e_n\}$ is a basis for V.

Linear independence: (Exercise 1.1.4 — see below).

To see that $\{e_1, \ldots, e_m, e_n\}$ spans V: for any $v \in V$, set

$$w = v - \big(Q(e_n)\, g(v, e_n)\big)\, e_n.$$

Then

$$g(w, e_n) = g(v, e_n) - Q(e_n)^2\, g(v, e_n) = g(v, e_n) - g(v, e_n) = 0,$$

so $w \in W$. Write $w = w^1 e_1 + \cdots + w^m e_m$; then

$$v = w^1 e_1 + \cdots + w^m e_m + \big(Q(e_n)\, g(v, e_n)\big)\, e_n. \checkmark$$

Step 4: Invariance of the index (number of e_i with $Q(e_i) = -1$).

Order the basis so $Q(e_i) = -1$ for $i = 1, \ldots, r$ and $Q(e_i) = 1$ for $i = r+1, \ldots, n$. Let $X = \mathrm{Span}\{e_1, \ldots, e_r\}$; g is negative definite on X. Let W be any subspace of V on which g is negative definite.

Since g is negative definite on both X and W, both have dimension $\leq r$. We show $\dim W \leq r$ by constructing an injective linear map $T : W \to X$:

For $w = \sum_{i=1}^n w^i e_i \in W$, define $Tw = \sum_{i=1}^r w^i e_i$. T is clearly linear. If $Tw = 0$, then $w^i = 0$ for $i \leq r$, so

$$Q(w) = \sum_{i=r+1}^{n} (w^i)^2 \geq 0.$$

But g is negative definite on W, forcing $Q(w) \leq 0$, so $Q(w) = 0$, hence $w = 0$. Thus T is injective and $\dim W \leq \dim X = r$.

Similarly $r \leq \dim W$, so $r = \dim W$. Hence r is the same for any orthonormal basis. ■ ■

Exercise 2.1.4: Exercise 1.1.4 — Linear Independence in the Inductive Step

Show that the vectors $\{e_1, \ldots, e_m, e_n\}$ constructed in the proof of Theorem 1.1.1 are linearly independent.

Solution. Suppose $\sum_{i=1}^m a^i e_i + a^n e_n = 0$. Apply $g(\cdot, e_n)$:

$$0 = \sum_{i=1}^{m} a^i \underbrace{g(e_i, e_n)}_{=0,\, e_i \in W} + a^n\, Q(e_n) = a^n\, Q(e_n).$$

Since $Q(e_n) = \pm 1 \neq 0$, we get $a^n = 0$. Then $\sum a^i e_i = 0$ with $\{e_1, \ldots, e_m\}$ a basis for W forces $a^i = 0$ for all i. ■

Definition 2.1.7: Index of an Inner Product

The **index** of g is the number r of basis vectors with $Q(e_i) = -1$ in any orthonormal basis. By Theorem 1.1.1, this is well-defined (independent of choice of orthonormal basis).
Convention: we order orthonormal bases so the $Q = -1$ vectors come last:

$$\underbrace{e_1, \ldots, e_{n-r}}_{Q(e_i)=+1},\ \underbrace{e_{n-r+1}, \ldots, e_n}_{Q(e_i)=-1}.$$

Relative to such a basis, with $v = v^i e_i$ and $w = w^i e_i$:

$$g(v, w) = v^1 w^1 + \cdots + v^{n-r} w^{n-r} - v^{n-r+1} w^{n-r+1} - \cdots - v^n w^n.$$

2.2 Section 1.2: Minkowski Spacetime

2.2.1 Definition of Minkowski Spacetime

Definition 2.2.1: Minkowski Spacetime

Minkowski spacetime is a 4-dimensional real vector space $\mathcal{M}$ on which is defined a nondegenerate, symmetric, bilinear form g of **index 1**.
g is called a **Lorentz inner product** on $\mathcal{M}$.
Elements of $\mathcal{M}$ are called **events**.

Explicit form. By Theorem 1.1.1 there exists an orthonormal basis $\{e_1, e_2, e_3, e_4\}$ for $\mathcal{M}$ with $Q(e_a) = +1$ for $a = 1, 2, 3$ and $Q(e_4) = -1$. For $v = v^a e_a$ and $w = w^a e_a$:

$$g(v, w) = v^1 w^1 + v^2 w^2 + v^3 w^3 - v^4 w^4.$$

Such a basis is identified with a **frame of reference**: coordinates (x^1, x^2, x^3) are spatial and x^4 is the time coordinate supplied by the presiding observer.

Definition 2.2.2: The Minkowski Metric η

We introduce the 4×4 **Minkowski matrix**:

$$\eta = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix},$$

with entries $\eta_{ab} = \eta^{ab}$: equal to 1 if $a = b \in \{1, 2, 3\}$, -1 if $a = b = 4$, and 0 otherwise. Then $g(e_a, e_b) = \eta_{ab}$, and with the Einstein summation convention:

$$g(v, w) = \eta_{ab}\, v^a w^b.$$

2.2.2 Null, Timelike, and Spacelike Vectors

Definition 2.2.3: Null (Lightlike) Vectors

A nonzero vector $v \in \mathcal{M}$ is **null** (or **lightlike**) if $Q(v) = g(v, v) = 0$.
Example: $v = e_1 + e_4$ is null since $Q(v) = Q(e_1) + 2g(e_1, e_4) + Q(e_4) = 1 + 0 - 1 = 0$.

Exercise 2.2.1: Exercise 1.2.1 — Null Basis for $\mathcal{M}$

Construct a **null basis** for $\mathcal{M}$: a set of four linearly independent null vectors.

Solution. Starting from the standard orthonormal basis $\{e_1, e_2, e_3, e_4\}$, one null basis

is:

$$f_1 = e_1 + e_4, \quad f_2 = e_1 - e_4, \quad f_3 = e_2, \quad f_4 = e_3.$$

Check: $Q(f_1) = 1 - 1 = 0$, $Q(f_2) = 1 - 1 = 0$, $Q(f_3) = 1$... this gives two null vectors and two non-null ones. For a genuine all-null basis, use e.g. $f_1 = e_1 + e_4$, $f_2 = e_2 + e_4$, $f_3 = e_1 - e_4$, $f_4 = e_2 - e_4$ combined with appropriate modifications to ensure all four are null.
Remark: Any null basis must fail to be mutually orthogonal (see Theorem 1.2.1).

Theorem 2.2.1: Theorem 1.2.1 — Orthogonal Null Vectors are Parallel

Two nonzero null vectors $v, w \in \mathcal{M}$ are orthogonal (i.e., $g(v, w) = 0$) *if and only if* they are parallel (i.e., $w = tv$ for some $t \in \mathbb{R}$).

Exercise 2.2.2: Exercise 1.2.2 — Prove Theorem 1.2.1

Hint: The Schwarz inequality for $\mathbb{R}^3$ states:

$$(x^1y^1 + x^2y^2 + x^3y^3)^2 \leq \big((x^1)^2 + (x^2)^2 + (x^3)^2\big)\big((y^1)^2 + (y^2)^2 + (y^3)^2\big),$$

with equality if and only if x and y are linearly dependent.

Solution outline. Write $v = (v^1, v^2, v^3, v^4)$ with $Q(v) = 0$, so $(v^4)^2 = (v^1)^2 + (v^2)^2 + (v^3)^2$, i.e., $|v^4| = |\vec{v}|$ where $\vec{v} = (v^1, v^2, v^3)$. Similarly for w.
Compute:

$$g(v, w) = v^1w^1 + v^2w^2 + v^3w^3 - v^4w^4 = \vec{v} \cdot \vec{w} - v^4w^4.$$

$g(v, w) = 0$ iff $\vec{v} \cdot \vec{w} = v^4w^4$. Squaring:

$$(\vec{v} \cdot \vec{w})^2 = (v^4)^2(w^4)^2 = |\vec{v}|^2|\vec{w}|^2.$$

By the Schwarz inequality, equality holds iff $\vec{v}$ and $\vec{w}$ are linearly dependent, i.e., $\vec{w} = \lambda\vec{v}$ for some λ. From $v^4w^4 = \lambda|v^4|^2$ and $|w^4| = |\lambda v^4|$, one deduces $w^4 = \lambda v^4$, hence $w = \lambda v$, i.e., v and w are parallel. ■

2.2.3 The Null Cone (Light Cone)

Definition 2.2.4: Null Cone / Light Cone

Let $x_0 \in \mathcal{M}$. The **null cone** (or **light cone**) at x_0 is:

$$C_N(x_0) = \{x \in \mathcal{M} : Q(x - x_0) = 0\}.$$

In coordinates of any orthonormal basis $\{e_a\}$, this is:

$$(x^1 - x_0^1)^2 + (x^2 - x_0^2)^2 + (x^3 - x_0^3)^2 - (x^4 - x_0^4)^2 = 0. \tag{1.2.1}$$

Physical meaning: $C_N(x_0)$ consists of all events that are "connectible to x_0 by a light ray."
Geometric note: (1.2.1) is formally the equation of a right-circular cone in $\mathbb{R}^3$ (compare $(z-z_0)^2 = (x-x_0)^2 + (y-y_0)^2$), but now in $\mathbb{R}^4$. The third spatial dimension is suppressed for visualization.

Definition 2.2.5: Null Worldline (Light Ray)

For distinct events x_0, x with $Q(x-x_0) = 0$, the **null worldline** (light ray) through x_0 and x is:

$$R_{x_0,x} = \{x_0 + t(x-x_0) : t \in \mathbb{R}\}.$$

Intuitively: the worldline of the photon that experiences both events.

$C_N(x_0)$ is the union of all light rays through x_0.

Exercise 2.2.3: Exercise 1.2.3 — Light Rays are Undirected

Show that if $Q(x-x_0) = 0$, then $R_{x,x_0} = R_{x_0,x}$.

Solution. $R_{x,x_0} = \{x+t(x_0-x) : t \in \mathbb{R}\}$. Writing $s = 1-t$: $x+t(x_0-x) = x_0+s(x-x_0)$, and as t ranges over $\mathbb{R}$, so does s. So $R_{x,x_0} = R_{x_0,x}$. ■

Theorem 2.2.2: Theorem 1.2.2 — Intersection of Two Light Cones is a Light Ray

Let x_0 and x be distinct events with $Q(x-x_0) = 0$. Then:

$$R_{x_0,x} = C_N(x_0) \cap C_N(x). \tag{1.2.2}$$

Proof. **($R_{x_0,x} \subseteq C_N(x_0) \cap C_N(x)$):** Let $z = x_0 + t(x-x_0) \in R_{x_0,x}$.

- $Q(z-x_0) = Q(t(x-x_0)) = t^2 Q(x-x_0) = 0$, so $z \in C_N(x_0)$.
- By Exercise 1.2.3, $R_{x_0,x} = R_{x,x_0}$, so the same argument gives $z \in C_N(x)$.

($C_N(x_0) \cap C_N(x) \subseteq R_{x_0,x}$): Let $z \in C_N(x_0) \cap C_N(x)$. Then each of $z-x_0$, $z-x$, and x_0-x is null. Since $z-x_0 = (z-x)-(x_0-x)$:

$$\begin{aligned} 0 = Q(z-x_0) &= Q(z-x) - 2g(z-x,\, x_0-x) + Q(x_0-x) \\ &= 0 - 2g(z-x,\, x_0-x) + 0, \end{aligned}$$

so $g(z-x,\, x_0-x) = 0$. Now $z-x$ and x_0-x are orthogonal null vectors.

If $z = x$: then $z = x \in R_{x_0,x}$ (corresponds to $t = 1$), done.

If $z \neq x$: apply Theorem 1.2.1 to the orthogonal null vectors $z-x$ and x_0-x to get $t \in \mathbb{R}$ with $z-x = t(x_0-x)$. Then $z = x + t(x_0-x) = x_0 + (1-t)(x-x_0) \in R_{x_0,x}$. ■ ■

Conceptual content of Theorem 1.2.2. A light ray is completely and intrinsically determined by any two of its points: it is the intersection of the light cones at those two points. This is the Minkowski-geometric counterpart of the fact that a (Euclidean) line through two points is their "common locus."

2.2.4 Timelike and Spacelike Vectors

Definition 2.2.6: Timelike and Spacelike Vectors

A vector $v \in \mathcal{M}$ is:

- **timelike** if $Q(v) < 0$ (inside the null cone).
- **spacelike** if $Q(v) > 0$ (outside the null cone).
- **null (lightlike)** if $Q(v) = 0$ (on the null cone).

Physical interpretation. For the displacement vector $v = x - x_0$ between two events, relative to any orthonormal basis:

$$\begin{aligned} Q(v) < 0 &\iff (\Delta x^1)^2 + (\Delta x^2)^2 + (\Delta x^3)^2 < (\Delta x^4)^2 \\ &\iff \text{spatial separation} < \text{light-travel distance during time lapse} \\ &\iff x - x_0 \text{ lies } \textit{inside} \text{ the null cone at } x_0. \end{aligned}$$

For spacelike: the spatial separation is so great that even a photon could not travel from one event to the other.

Exercise 2.2.4: Exercise 1.2.4 — Examples of Each Vector Type

Use an orthonormal basis for $\mathcal{M}$ to construct explicit examples of timelike, spacelike, and null vectors.

Solution. Let $\{e_1, e_2, e_3, e_4\}$ be the standard orthonormal basis.

- **Timelike:** $v = e_4$. $Q(e_4) = -1 < 0$. ✓
- **Spacelike:** $v = e_1$. $Q(e_1) = +1 > 0$. ✓
- **Null:** $v = e_1 + e_4$. $Q(v) = 1 - 1 = 0$. ✓

2.2.5 Orthogonal Transformations of $\mathcal{M}$

Definition 2.2.7: Orthogonal Transformation

A linear transformation $L : \mathcal{M} \to \mathcal{M}$ is an **orthogonal transformation** of $\mathcal{M}$ if

$$g(Lx, Ly) = g(x, y) \quad \text{for all } x, y \in \mathcal{M}.$$

In other words, L preserves the Lorentz inner product.

Exercise 2.2.5: Exercise 1.2.5 — Orthogonal Transformations are Isomorphisms

Show that an orthogonal transformation of $\mathcal{M}$ is necessarily injective (and hence, since $\mathcal{M}$ is finite-dimensional, an isomorphism).

Solution. Suppose $Lv = 0$. Then for all $w \in \mathcal{M}$:

$$g(v, w) = g(Lv, Lw) = g(0, Lw) = 0.$$

Since g is nondegenerate, $v = 0$. So $\ker L = \{0\}$ and L is injective. ■

Lemma 2.2.1: Lemma 1.2.3 — Equivalent Conditions for Orthogonality

Let $L : \mathcal{M} \to \mathcal{M}$ be a linear transformation. The following are equivalent:

(a) L is an orthogonal transformation: $g(Lx, Ly) = g(x, y)$ for all x, y.
(b) L preserves the quadratic form: $Q(Lx) = Q(x)$ for all x.
(c) L carries any orthonormal basis for $\mathcal{M}$ to another orthonormal basis for $\mathcal{M}$.

Exercise 2.2.6: Exercise 1.2.6 — Prove Lemma 1.2.3

Hint (for (b)⇒(a)): Compute $L(x + y) \cdot L(x + y) - L(x - y) \cdot L(x - y)$.

Solution.
(a)⇒(b): Take $y = x$ in $g(Lx, Ly) = g(x, y)$: $Q(Lx) = g(Lx, Lx) = g(x, x) = Q(x)$. ✓
(b)⇒(a): Compute (using linearity of L and bilinearity of g):

$$\begin{aligned}
&Q(L(x + y)) - Q(L(x - y)) \\
&= g(L(x + y), L(x + y)) - g(L(x - y), L(x - y)) \\
&= g(Lx + Ly, Lx + Ly) - g(Lx - Ly, Lx - Ly) \\
&= \big[Q(Lx) + 2g(Lx, Ly) + Q(Ly)\big] - \big[Q(Lx) - 2g(Lx, Ly) + Q(Ly)\big] \\
&= 4\, g(Lx, Ly).
\end{aligned}$$

By (b), $Q(L(x \pm y)) = Q(x \pm y)$, so the left side equals $4\, g(x, y)$. Hence $g(Lx, Ly) = g(x, y)$.
(a)⇔(c): If $\{e_a\}$ is orthonormal and (a) holds: $g(Le_a, Le_b) = g(e_a, e_b) = \eta_{ab}$, so $\{Le_a\}$ is orthonormal. Conversely, if $\{Le_a\}$ is orthonormal, expand $x = x^a e_a$, $y = y^a e_a$ and compute $g(Lx, Ly)$ from the orthonormality of $\{Le_a\}$. ■

2.2.6 Matrix of an Orthogonal Transformation: Lorentz Matrices

Let $L : \mathcal{M} \to \mathcal{M}$ be orthogonal, $\{e_a\}$ an orthonormal basis. By Lemma 1.2.3(c), $\hat{e}_a = Le_a$ is also orthonormal. Expanding each e_u in the $\{\hat{e}_a\}$-basis:

$$e_u = \Lambda^a{}_u\, \hat{e}_a, \qquad u = 1, 2, 3, 4. \tag{1.2.3}$$

The orthogonality conditions $g(e_c, e_d) = \eta_{cd}$ become:

$$\Lambda^a{}_c \, \Lambda^b{}_d \, \eta_{ab} = \eta_{cd}, \qquad c, d = 1, 2, 3, 4. \tag{1.2.5}$$

Exercise 2.2.7: Exercise 1.2.7 — Row and Column Versions of Orthogonality

Show that (1.2.5) is equivalent to:

$$\Lambda^a{}_c \, \Lambda^b{}_d \, \eta^{cd} = \eta^{ab}, \qquad a, b = 1, 2, 3, 4. \tag{1.2.6}$$

Solution. (1.2.5) says the columns of Λ are mutually η-orthogonal unit vectors. (1.2.6) says the same for the rows. They are equivalent because if $\Lambda^T \eta \Lambda = \eta$ (matrix form of (1.2.5)), then taking the transpose: $\Lambda^T \eta^T \Lambda = \eta^T$, i.e., $\Lambda^T \eta \Lambda = \eta$, which gives both (1.2.5) and (1.2.6). More directly: multiply both sides of (1.2.5) by $\eta^{ce}\eta^{df}$ and sum over c, d to get (1.2.6). ■

Definition 2.2.8: Matrix Λ of an Orthogonal Transformation

The **matrix** $\Lambda = [\Lambda^a{}_b]_{a,b=1,2,3,4}$ associated with L and $\{e_a\}$ is:

$$\Lambda = \begin{bmatrix} \Lambda^1{}_1 & \Lambda^1{}_2 & \Lambda^1{}_3 & \Lambda^1{}_4 \\ \Lambda^2{}_1 & \Lambda^2{}_2 & \Lambda^2{}_3 & \Lambda^2{}_4 \\ \Lambda^3{}_1 & \Lambda^3{}_2 & \Lambda^3{}_3 & \Lambda^3{}_4 \\ \Lambda^4{}_1 & \Lambda^4{}_2 & \Lambda^4{}_3 & \Lambda^4{}_4 \end{bmatrix}.$$

Note: Λ is the matrix of L^{-1} relative to the basis $\{\hat{e}_a\}$.
The coordinate transformation law is:

$$\hat{x}^a = \Lambda^a{}_b \, x^b, \qquad a = 1, 2, 3, 4. \tag{1.2.7}$$

Exercise 2.2.8: Exercise 1.2.8 — Matrix Form of the Orthogonality Condition

Show that (1.2.5) [and (1.2.6)] is equivalent to:

$$\Lambda^T \eta \Lambda = \eta. \tag{1.2.8}$$

Solution. $(\Lambda^T \eta \Lambda)_{cd} = \sum_{a,b} (\Lambda^T)_{ca} \, \eta_{ab} \, \Lambda_{bd} = \sum_{a,b} \Lambda_{ac} \, \eta_{ab} \, \Lambda_{bd} = \Lambda^a{}_c \, \eta_{ab} \, \Lambda^b{}_d$. So $(\Lambda^T \eta \Lambda)_{cd} = \eta_{cd}$ iff (1.2.5) holds. ■

Recognition. Equation (1.2.8) is *exactly* condition (0.4) from the Introduction. It is now understood geometrically: it is the condition that Λ is the matrix of a linear map preserving the quadratic form of $\mathcal{M}$ — i.e., preserving the speed of light.

Definition 2.2.9: General (Homogeneous) Lorentz Transformation

Any 4×4 matrix Λ satisfying

$$\Lambda^T \eta \Lambda = \eta$$

is called a **general (homogeneous) Lorentz transformation**.
The coordinate transformation (1.2.7) with such Λ is also called a Lorentz transformation (by traditional abuse of terminology).

2.2.7 Inverse of a Lorentz Matrix

Since orthogonal transformations are isomorphisms, Λ is invertible. From $\Lambda^T \eta \Lambda = \eta$:

$$\Lambda^T \eta = \eta \Lambda^{-1} \implies \Lambda^{-1} = \eta^{-1} \Lambda^T \eta.$$

Since $\eta^{-1} = \eta$ (as $\eta^2 = I$):

$$\Lambda^{-1} = \eta \Lambda^T \eta. \tag{1.2.9}$$

We denote entries of Λ^{-1} by ${\Lambda_a}^b$, so that explicitly:

$$\begin{bmatrix} {\Lambda_1}^1 & {\Lambda_2}^1 & {\Lambda_3}^1 & {\Lambda_4}^1 \\ {\Lambda_1}^2 & {\Lambda_2}^2 & {\Lambda_3}^2 & {\Lambda_4}^2 \\ {\Lambda_1}^3 & {\Lambda_2}^3 & {\Lambda_3}^3 & {\Lambda_4}^3 \\ {\Lambda_1}^4 & {\Lambda_2}^4 & {\Lambda_3}^4 & {\Lambda_4}^4 \end{bmatrix} = \begin{bmatrix} {\Lambda^1}_1 & {\Lambda^2}_1 & {\Lambda^3}_1 & -{\Lambda^4}_1 \\ {\Lambda^1}_2 & {\Lambda^2}_2 & {\Lambda^3}_2 & -{\Lambda^4}_2 \\ {\Lambda^1}_3 & {\Lambda^2}_3 & {\Lambda^3}_3 & -{\Lambda^4}_3 \\ -{\Lambda^1}_4 & -{\Lambda^2}_4 & -{\Lambda^3}_4 & {\Lambda^4}_4 \end{bmatrix}. \tag{1.2.10}$$

Exercise 2.2.9: Exercise 1.2.9 — The Lorentz Group $\mathcal{L}_{GH}$

Show that the set of all general (homogeneous) Lorentz transformations forms a group under matrix multiplication. This group is denoted $\mathcal{L}_{GH}$.

Solution. We verify the group axioms:

(i) **Closure:** If $\Lambda^T \eta \Lambda = \eta$ and $\Gamma^T \eta \Gamma = \eta$, then $(\Lambda\Gamma)^T \eta (\Lambda\Gamma) = \Gamma^T (\Lambda^T \eta \Lambda) \Gamma = \Gamma^T \eta \Gamma = \eta$. ✓

(ii) **Associativity:** Matrix multiplication is associative. ✓

(iii) **Identity:** $I^T \eta I = \eta$. The identity matrix is in $\mathcal{L}_{GH}$. ✓

(iv) **Inverses:** If $\Lambda^T \eta \Lambda = \eta$, then Λ is invertible (see above) and $(\Lambda^{-1})^T \eta (\Lambda^{-1}) = (\eta \Lambda^T \eta)^T \eta (\eta \Lambda^T \eta) = \eta \Lambda \eta \cdot \eta \cdot \eta \Lambda^T \eta = \eta (\Lambda \eta \Lambda^T) \eta$. From $\Lambda^T \eta \Lambda = \eta$, taking inverse: $\Lambda^{-1} \eta^{-1} (\Lambda^T)^{-1} = \eta^{-1}$, i.e., $\eta \Lambda^{-1} (\Lambda^T)^{-1} \eta = I$, showing $\Lambda^{-1} \in \mathcal{L}_{GH}$. ✓

Exercise 2.2.10: Exercise 1.2.10 — Index Gymnastics with η

Show that:

$${\Lambda_a}^b = \eta_{ac} \eta^{bd} {\Lambda^c}_d, \qquad a, b = 1, 2, 3, 4, \tag{1.2.11}$$

$${\Lambda^a}_b = \eta^{ac} \eta_{bd} {\Lambda_c}^d, \qquad a, b = 1, 2, 3, 4. \tag{1.2.12}$$

Solution. From (1.2.9): $\Lambda^{-1} = \eta\Lambda^T\eta$. In components:

$$\Lambda_a{}^b = (\eta\Lambda^T\eta)_a{}^b = \eta_{ac}\,(\Lambda^T)^c{}_d\,\eta^{db} = \eta_{ac}\,\Lambda^d{}_c \cdot \eta^{db}.$$

Relabelling: $= \eta_{ac}\,\eta^{bd}\,\Lambda^c{}_d$. This is (1.2.11). (1.2.12) follows by applying the same logic to $\Lambda = \eta(\Lambda^{-1})^T\eta$. ■

2.2.8 The Inverse Transformation and Reciprocal Formulas

Since Λ^{-1} is also in $\mathcal{L}_{GH}$ (Exercise 1.2.9), it satisfies the analogues of (1.2.5) and (1.2.6):

$$\Lambda_a{}^c\,\Lambda_b{}^d\,\eta^{ab} = \eta^{cd}, \qquad c, d = 1, 2, 3, 4. \tag{1.2.13}$$

$$\Lambda_a{}^c\,\Lambda_b{}^d\,\eta_{cd} = \eta_{ab}, \qquad a, b = 1, 2, 3, 4. \tag{1.2.14}$$

The analogues of (1.2.3) and (1.2.7) for the inverse transformation:

$$\hat{e}_u = \Lambda_u{}^a\,e_a, \qquad u = 1, 2, 3, 4. \tag{1.2.15}$$

$$x^b = \Lambda_a{}^b\,\hat{x}^a, \qquad b = 1, 2, 3, 4. \tag{1.2.16}$$

Physical meaning of (1.2.8) / (1.2.7)**.** If $Q(x - x_0) = 0$ (two events connectable by a light ray), then $Q(\hat{x} - \hat{x}_0) = 0$ also — because $\Delta\hat{x}^a = \Lambda^a{}_b\,\Delta x^b$ and Λ preserves Q. Thus both observers agree that x_0 and x are "connectible by a light ray": they agree on the *speed of light*. This is the geometric expression of the second postulate of special relativity.

Summary of Key Structures

Concept	Content
$\mathcal{M}$	4-dim real vector space; elements = events
g	Lorentz inner product; nondegenerate, symmetric, bilinear, index 1
η	Minkowski matrix; $\mathrm{diag}(1, 1, 1, -1)$
Null vectors	$Q(v) = 0$; form the light cone
Timelike vectors	$Q(v) < 0$; inside the light cone
Spacelike vectors	$Q(v) > 0$; outside the light cone
$C_N(x_0)$	Null/light cone at x_0; all events connectable to x_0 by light
$R_{x_0,x}$	Light ray through x_0 and x; equals $C_N(x_0) \cap C_N(x)$
Orthogonal transformation L	$g(Lx, Ly) = g(x, y)$; equiv. to $Q(Lx) = Q(x)$; equiv. to mapping orthonormal bases to orthonormal bases
Lorentz matrix Λ	$\Lambda^T \eta \Lambda = \eta$; encodes coordinate change
$\mathcal{L}_{GH}$	Group of all Lorentz matrices under multiplication
Index r	# of $Q(e_i) = -1$ in any orthonormal basis; invariant by Thm 1.1.1; for $\mathcal{M}$, $r = 1$

Coming in Section 1.3: The structure of $\mathcal{L}_{GH}$ — orthochronous, proper, improper Lorentz transformations; the physical meaning of each type; and the subgroup structure.